Core Concepts:

Kubernetes: Container orchestration
Cluster: Group of nodes
Node: Machine
Master Node / Control Plane: Management layer
Worker Node: Executor
API Server: Gateway(communication hub)
Scheduler: Planner
Controller Manager: Supervisor
ETCD: key-value database
Kubelet: Agent
Kube-Proxy: Router
Container Runtime: Engine

Workload Objects:

Pod: Smallest deployable unit
ReplicaSet: Replication(Desired = Actual)
Deployment: Updater(ReplicaSet and provides updates to pod)
StatefulSet: Stateful(Everything is stable)
DaemonSet: Node-service
Job: Task
CronJob: Scheduler

Networking Concepts:

Service: Access(permanent address)
ClusterIP: Internal
NodePort: External
LoadBalancer: Distributor
Ingress: Rules(Instructions where to send traffic)
Ingress Controller: Traffic-manager
DNS: Converts names (domain/service names) into IP addresses.
Network Policies: Firewall
CNI Plugins: Networking for pods

Storage Concepts:

Volume: Storage
Persistent Volume (PV): Disk
Persistent Volume Claim (PVC): Request
Storage Class: Storage type/template.
Dynamic Provisioning: Automatic
ConfigMap: Configuration
Secret: Sensitive
CSI: bridge between Kubernetes and storage systems

Configuration & Management:

Labels: Tags
Selectors: Filters
Annotations: Extra information of Metadata
Namespaces: Isolation
Resource Quotas: Limits
Limit Ranges: Boundaries
Taints: Restriction
Tolerations: Permission
Affinity: Preference
Anti-Affinity: Reverse of Affinity
Node Selector: Selection

Security Concepts:

RBAC: Authorization
Service Account: Pod Identity
Role: Permission only Namespace level
RoleBinding: bridge between user and role only Namespace level
ClusterRole: Global-permission
ClusterRoleBinding: bridge between user and role only cluster
Pod Security Standards: Namespace/cluster security
Security Context: Pod/Container security
TLS Certificates: Encryption(https)